





About Us



- **Bridging Farmers Together:** A platform where small & needy farmers can access tractors, harvesters, land, and labor from big farmers.
- **Resource Sharing Economy:** Enables cost-effective access to modern farming equipment and manpower.
- **AI & ML Integration:**
 - Crop prediction, price prediction & yield forecasting
 - Smart suggestions on which crops to grow based on soil, weather, and market trends.
- **Farmer-Friendly Assistant Bot:** Provides instant answers, and guidance regarding any service.
- **All-in-One Agri Platform:** From equipment booking to crop planning – everything at one place.
- **Mission:** Empower farmers, reduce costs, and boost agricultural productivity.

MODULES

A rectangular card with rounded corners. The top half has a light blue background with white clouds. The bottom half has a green background with rolling hills. The text is centered in the blue area.

**User Accounts &
Dashboards**

A rectangular card with rounded corners. The top half has a light blue background with white clouds. The bottom half has a green background with rolling hills. The text is centered in the blue area.

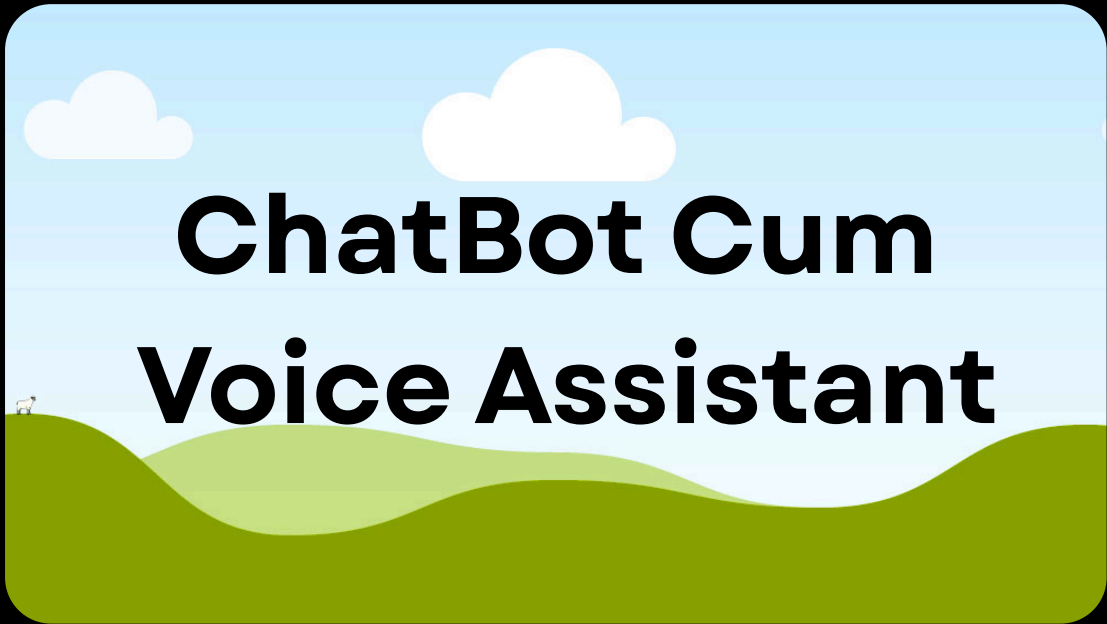
**Lending & Equipment
Management**

A rectangular card with rounded corners. The top half has a light blue background with white clouds. The bottom half has a green background with rolling hills. The text is centered in the blue area.

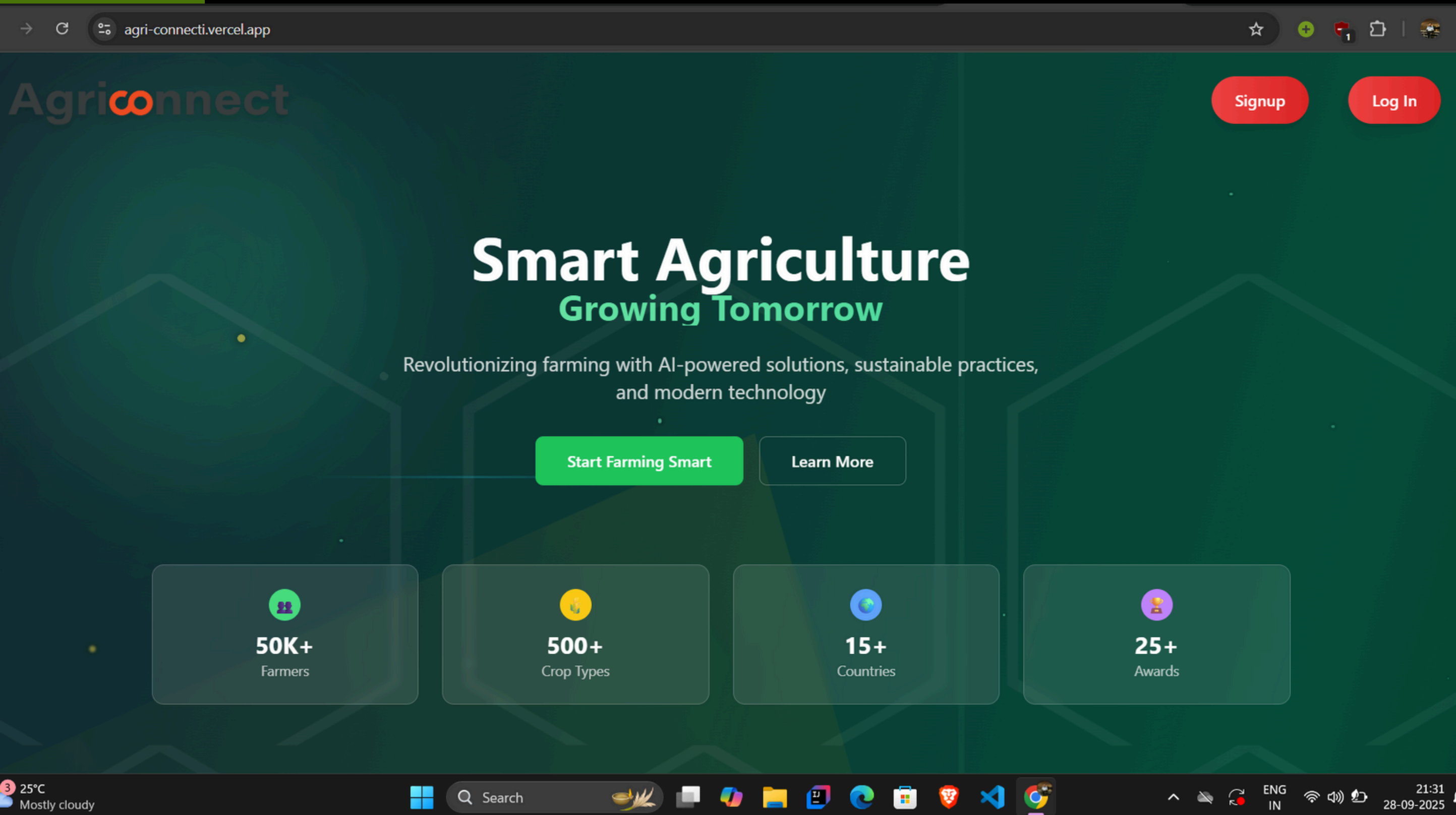
Booking System

A rectangular card with rounded corners. The top half has a light blue background with white clouds. The bottom half has a green background with rolling hills. The text is centered in the blue area.

**AI-powered
Assistance**

A rectangular card with rounded corners. The top half has a light blue background with white clouds. The bottom half has a green background with rolling hills. The text is centered in the blue area.

**ChatBot Cum
Voice Assistant**



Home Page

03

create account

Login



A Join AgriConnect

Create your account and start your smart farming journey

Personal Information

Full Name

Enter your full name

Email Address

farmer@example.com

Phone Number

+91 9876543210

Security

Password

Create a strong password

Location Details

Country

India

State

e.g., Maharashtra

Pincode

123456

Address

Street, City

☐ I agree to the [Terms of Service](#) and [Privacy Policy](#)

Create Account >

Already have an account? [Sign in here](#)

Join 10,000+ Farmers

Start your smart farming journey today!



Equipment Rental

Access 500+ modern farming equipment at your fingertips



AI-Powered Insights

Get personalized farming recommendations and market trends



Smart Analytics

Track your farm performance with data-driven insights



50+ Cities

Nationwide network with 24/7 support coverage

Trusted by farmers across India

4.9★ 99% 24/7
Rating Uptime Support

Agriconnect

AgriConnect

Smart Farming Solutions



Equipment Rental

Access modern farming equipment at affordable prices



AI Assistant

Get smart farming guidance and insights



Welcome Back!

Enter your credentials to access your farming dashboard

Email Address

farmer@example.com

Password

Enter your password

[Forgot password?](#)

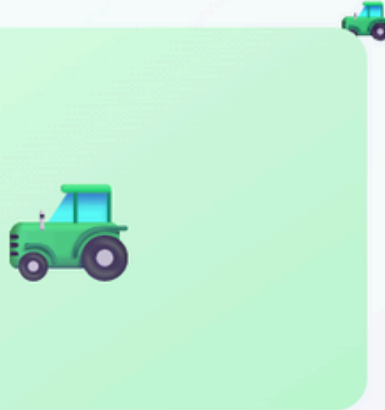
Login to Dashboard >

or

New to AgriConnect? [Create an account](#)

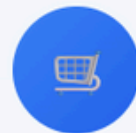
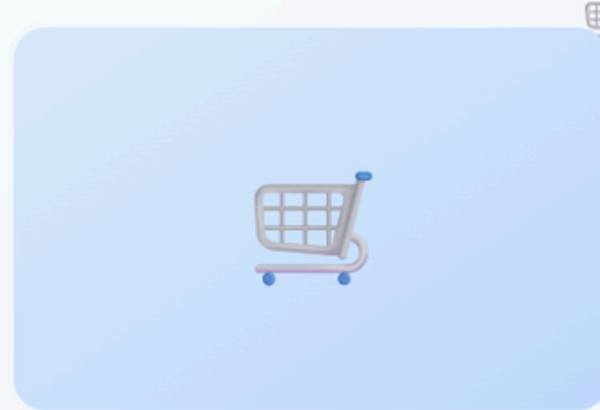
Welcome to AGRICONNECT

Revolutionizing agriculture with smart technology solutions for modern farmers



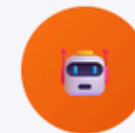
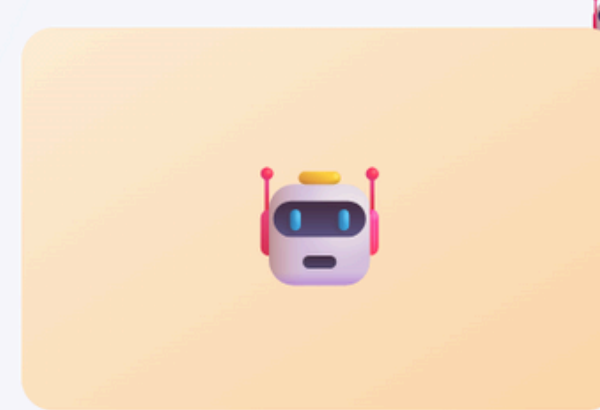
Lend Equipment

Access modern farming equipment at affordable prices. Boost your productivity without high upfront costs.



Rent Products

Purchase quality agricultural tools, seeds, and machinery from trusted sellers with complete confidence.



AI Assistant

Get AI-powered farming guidance, weather forecasts, and crop care tips for maximum yields.

dashboard

Agriconnect

What would you like to lend today?

Choose from our wide range of agricultural equipment and services

**Tractor**

Heavy-duty tractors for all farming needs

[Lend Now](#)**HARVESTOR**

Efficient harvesting equipment

[Lend Now](#)**Labour Services**

Skilled agricultural workers

[Lend Now](#)

List Tractor for Rent

Fill out the details below

Title *

Tractor

Description *

Heavy-duty tractors for all farming needs

Price *

Enter price

Pricing Unit *

Select pricing unit

Category *

Tractor

Country *

Enter country

State *

Enter state

Lending Page

03

Our Services



Available Harvester

Choose from our verified providers

DESCRIPTION	PRICE	PRICE TYPE	ACTION
Efficient harvesting equipment	₹1200	PER_DAY	<button>Rent Now</button>
Efficient harvesting equipment	₹1200	PER_DAY	<button>Rent Now</button>
good powerfull	₹1200	PER_DAY	<button>Rent Now</button>

BOOKING

agri-connecti.vercel.app says

✓ Booking confirmed! Your requested Service will reach to your address within 30 mins. See Profile for booking details

OK

Booking Confirmation

User Profile

A

Adarsh Dubey

My Profile

My Bookings

Requested Services

Log Out

A

Adarsh Dubey

Email

adarsh@gmail.com

Phone Number

9359976714

Country

India

State

Kerala

Pincode

6866635

Address

IIIT KOTTAYAM CAMPUS,VALAVOOR

My Bookings



HARVESTOR

 9/28/2025  4:12:39 PM  B11-cooptyre(palazhy)hostel,IIIT Kottayam valavoor, Kerala

COMPLETED**Booking Completed**

HARVESTOR

 9/17/2025  3:21:01 PM  IIIT KOTTAYAM CAMPUS,VALAVOOR, Kerala

BOOKED**Cancel**

Tractor

 9/15/2025  6:20:49 PM  IIIT KOTTAYAM CAMPUS,VALAVOOR, Kerala

BOOKED**Cancel**

HARVESTOR

 9/11/2025  6:51:42 PM  IIIT KOTTAYAM CAMPUS,VALAVOOR, Kerala

COMPLETED**Booking Completed**

Tractor

 9/9/2025  7:59:29 PM  B11-cooptyre(palazhy)hostel,IIIT Kottayam valavoor, Kerala

CANCELLED**Booking Cancelled**

My Booking

Requested Services



HARVESTOR

Requested by: Adarsh Dubey

📅 9/28/2025 ⌚ 4:12:39 PM

COMPLETED

₹1200



HARVESTOR

Requested by: Adarsh Dubey

📅 9/17/2025 ⌚ 3:21:01 PM

BOOKED

₹1000

Mark as Completed



Tractor

Requested by: Adarsh Dubey

📅 9/15/2025 ⌚ 6:20:49 PM

BOOKED

₹11199.99

Mark as Completed



HARVESTOR

Requested by: Adarsh Dubey

📅 9/11/2025 ⌚ 6:51:42 PM

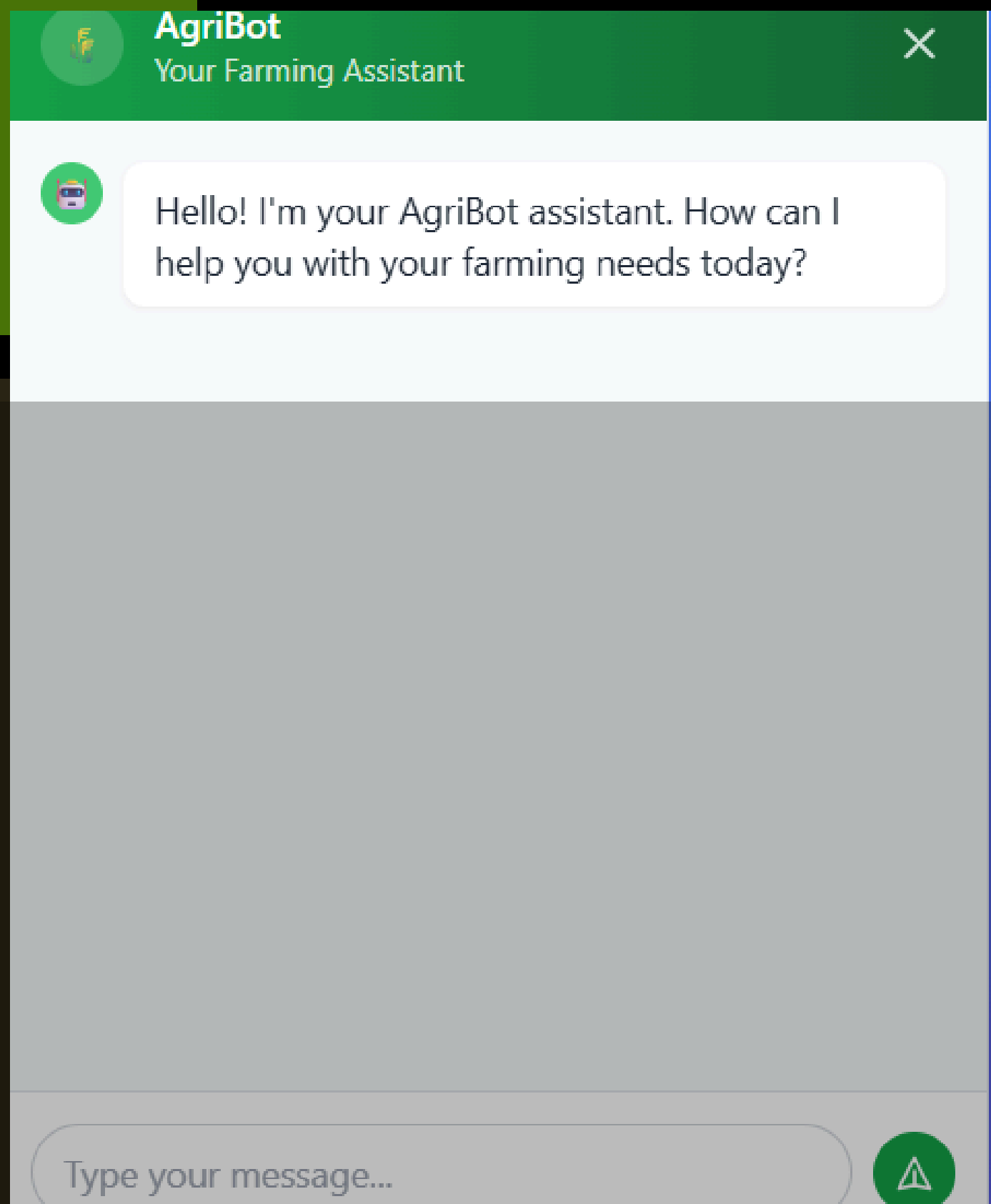
COMPLETED

₹1000

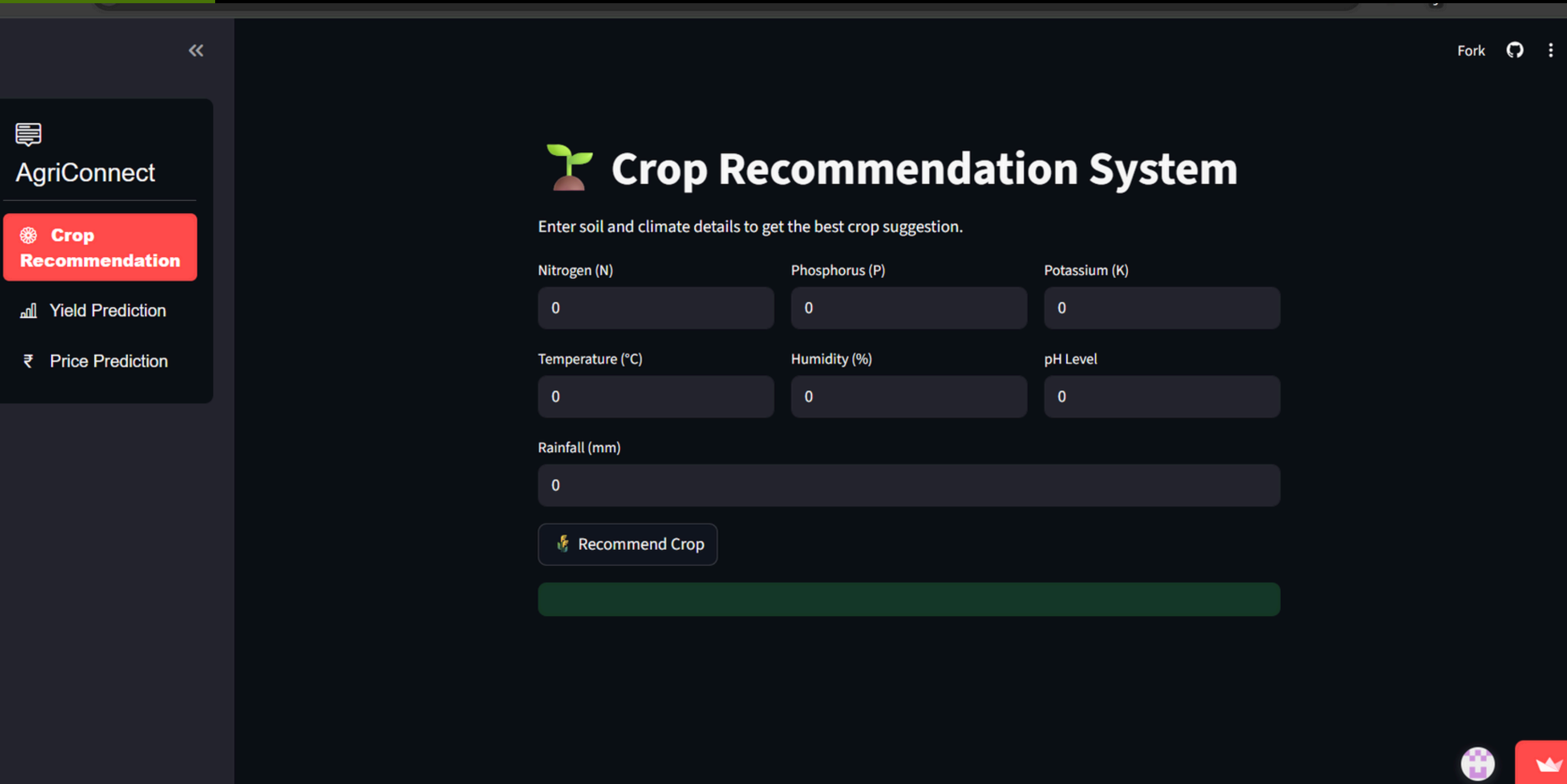
List of your
equipment
requested

03

Our Services



ChatBot cum Voice Assistant



AI BASED ASSISTANT

Agile Methodology.

Iterative and incremental development → allows continuous feedback, quick feature delivery, and adaptability to farmer needs.

Abstraction & Decomposition

Complex problems (crop prediction, booking system, chatbot) are broken into smaller, manageable modules for easier development and testing.

Microservices-Oriented Design

Each core function (Authentication, Equipment Booking, Crop Prediction, Chatbot) is designed as independent, loosely coupled services → scalable and maintainable.

Separation of Concerns

Clear division between frontend, backend, and database layers ensures modularity and reduces complexity.

Cloud-Native Approach

Designed to leverage containerization (Docker) and cloud hosting (Render + Neon DB) for scalability, resilience, and easy deployment.



- ✓ Spring Boot Framework for building scalable and modular backend services.
- 🔒 JWT-based Authentication with Spring Security to protect farmer and vendor data.
- ⚡ RESTful APIs for smooth communication between frontend, mobile, and backend.
- 🐳 Dockerized Deployment ensuring consistency across environments.
- ☁ Hosted on Render Cloud for high availability and auto-scaling.
- 🗄 PostgreSQL (Neon Cloud) as a secure, cloud-native database for storing farmers, equipment, bookings, and crop insights.
- 🔧 Future-Ready Microservice Design to easily integrate ML crop predictor & AI chatbot.

Why These Technologies?

Technology	Why We Chose It
 Spring Boot	- Rapid development with minimal boilerplate. - Production-ready features (monitoring, embedded servers). - Scalable & modular, future-ready for ML & chatbot integration.
 Spring Security + JWT	- Industry-standard authentication & authorization. - Stateless, token-based authentication with JWT. - Ensures secure access & role-based permissions.
 Docker & Containerization	- Packages app + dependencies in portable containers. - Guarantees consistency across dev, test, and prod. - Simplifies scaling & microservice adoption.
 Render Cloud + Neon DB	- Managed hosting with auto-scaling & high availability. - Easy CI/CD integration for smooth deployments. - Neon PostgreSQL (Cloud-native DB) ensures reliable, scalable, and secure data storage. - Unified cloud ecosystem for backend + database.
 Redux (Frontend State Management)	- Provides a centralized store for managing app state. - Ensures smooth synchronization between frontend & backend APIs. - Improves scalability of frontend by reducing complexity in state handling.

Voice + text assistant chatbot- *Technologies Used*

Voice to Text / Text to Text AI Assistant with RAG (Retrieval-Augmented Generation)

- FastAPI – Backend framework to handle APIs (speech, text, file upload).
- Whisper (Faster-Whisper) – Converts speech → text (STT).
- LangChain + LangGraph – Orchestration for RAG pipeline & chat flow.
- Pinecone – Vector database for storing and retrieving embeddings.
- FAISS (optional local) – Alternative vector search for PDFs.
- HuggingFace Embeddings – Converts text into vectors.
- Groq LLM (LLaMA 3) – Generates answers based on context & history.
- SQLite – Stores chat history for personalization.
- PDF Processing (PyPDF2 + Text Splitter) – For knowledge ingestion.

Why These & Benefits

Why Used?



FastAPI → Lightweight, fast, integrates well with frontend.

Whisper → Accurate multilingual speech recognition.

LangChain + Pinecone → Enables RAG → context-aware, factual answers.

Groq LLM → Efficient, scalable, cost-effective AI responses.

-  Voice-based interaction – More natural than typing.
-  Text-to-text chat – Flexible input for different users.
- ⚡ Fast responses – Optimized with CPU-friendly Whisper + Groq.

Smart Crop Recommendation

"Identify the best-suited crop for your land and environment to ensure higher productivity and sustainable farming."

Crop Recommendation System

Enter soil and climate details to get the best crop suggestion.

Nitrogen (N)	Phosphorus (P)	Potassium (K)
0	0	0
Temperature (°C)	Humidity (%)	pH Level
0	0	0
Rainfall (mm)		
0		

 Recommend Crop

Intelligent Yield Prediction

"Estimate crop yield based on soil, climate, and farming inputs to plan cultivation with confidence and efficiency."

Yield Prediction using ML

Enter crop, season, and environmental details to estimate yield per hectare.

Crop	Season	State
Select Crop	Select Season	Select State
Crop Year (YYYY)	Area (hectares)	Production (tonnes)
Annual Rainfall (mm)	Fertilizer Usage (kg)	Pesticide Usage (kg)
Distance from places		
4		
Predict Yield		

Smart Crop Recommendation

"Predict crop prices with data-driven insights to help farmers make informed selling decisions and maximize profits."

Crop Price Prediction using ML

Enter the crop and environmental details to estimate market price.

Crop	Season	State
-- Select Crop --	-- Select Season --	-- Select State --
Cultivated Area (hectares)	Soil Type	Pesticide Usage
0.00	-- Select Soil Type --	-- Select Pesticide Usa...
Soil pH Level	Temperature	Fertilizer Usage
-- Select pH Level --	-- Select Temperature --	-- Select Fertilizer Usa...
Annual Rainfall (mm)		
0.00		

Predict Price

Frontend Architecture for AgriConnect

UI/UX Considerations

- **Responsive Design:** Fully adaptable across desktop, tablet, and mobile devices using Tailwind's responsive utilities.
- **Accessibility:** Semantic HTML elements, keyboard navigation support, and ARIA labels.
- **Performance Optimization:** Code-splitting, lazy loading of components, and optimized asset delivery.
- **Motion & Animation:** Smooth transitions and feedback using React Motion to improve engagement without overwhelming users.
- **Consistent Theme:** Custom Tailwind config ensures consistent color palette, fonts, and spacing throughout the platform.

Key Technologies

- **React:** Component-based architecture for reusable UI elements, state management, and dynamic rendering.
- **Tailwind CSS:** Utility-first CSS framework for rapid styling and responsiveness across devices.
- **HTML5 & CSS3:** Base markup and styling for semantic and accessible structure.
- **React Motion / Framer Motion:** For smooth animations and interactive motion effects.
- **JavaScript ES6+:** Logic and interactivity throughout the application.

Loose Coupling in AgriConnect

Independent Communication – Modules interact through APIs or message queues, not direct code dependencies.

Interchangeability – Any module (e.g., AI Crop Prediction) can be upgraded or replaced without breaking others.

Minimal Knowledge – Each module only knows what service another provides, not how it works internally.

Scalability – Modules can scale independently (e.g., Booking may scale for peak seasons without affecting Dashboards).

Resilience & Flexibility – If one module fails or changes, the rest continue to function with minimal impact.

Coupling

High Coupling (Bad):

- Booking System ↔ User Accounts & Dashboards → tightly linked (you can't book without being logged in).
- Booking System ↔ Lending & Equipment Management → booking depends on equipment data.

Low Coupling (Good):

- AI-powered Assistance → can work independently (just needs data inputs).
- ChatBot Cum Voice Assistant → loosely coupled, mainly interacts via APIs to fetch info.

Cohesion

High Cohesion (Good):

- User Accounts & Dashboards → all tasks are related to users (login, role-based dashboards).
- Booking System → purely booking-related tasks.

Lower Cohesion (Weaker):

- ChatBot Cum Voice Assistant → mixes two concerns (website navigation help + agri queries). Could be split for higher cohesion.
- AI-powered Assistance → if it handles both prediction + recommendations + trends, cohesion might drop unless it is tightly scoped to “crop prediction & yield.”

Thank
You!

